

PROBLEM & OPPORTUNITY

Bangladesh carries a **double burden of malnutrition** — 24% child stunting, 11% adult diabetes, and rapidly rising obesity in women (now 35%, up from 11% in 2004). Hypertension and anaemia compound the crisis. Yet every major nutrition app is English-only, Western-food-centric, and functionally useless for 170 million people who eat hilsa, khichuri, and dal as daily staples. NutriDesh closes this gap.

24%

Child stunting nationally

11%

Adults with diabetes

35%

Women overweight/obese

~50

BD foods currently offline

CORE FEATURES (MVP)

AI Food Scanner — Scan & Know

Scan raw bazar ingredients or a cooked deshi plate. Returns a nutrition breakdown (calories, protein, iron, calcium, vitamin A) using a curated Bangladeshi food dataset and Gemini vision analysis.

Bangla AI Nutrition Chat

Conversational assistant in Bangla. Knows the user's profile, medical conditions, and day's logged meals. Answers contextual questions like *"Diabetes thakle ilish mach khaoa thik?"*

Daily Nutrition Dashboard

Thali progress ring, macro bars, and key micronutrients against Bangladeshi RDA values. Logging via camera scan or manual search. Deficiency alerts for at-risk nutrition patterns.

Diabetes Focus Mode

Glycemic load guidance, traffic-light food flags (green / yellow / red), and blood glucose logging — helping users connect Bangladeshi meals directly to sugar control.

AI & TECHNOLOGY STACK

A unified **Gemini Flash** engine powers image recognition, Bangla chat, and recommendation generation — entirely within the free tier with zero paid API cost. Offline caching currently stores frequently used Bangladeshi foods locally using AsyncStorage, while Gemini handles dynamic analysis and conversation.

Expo React Native

Node.js / Express

Render (free tier)

AsyncStorage

Gemini Flash

LOCAL DIFFERENTIATION

NutriDesh is built Bangladeshi from the ground up — not a Western tracker with Bangla translations bolted on. The food dataset, prompts, UI copy, and health modes are grounded in Bangladeshi dishes, Bangla language, and clinical nutrition references from **BIRDEM, FAO, and WHO**. This makes it a nutrition companion that genuinely understands how Bangladeshis eat.

HACKATHON DELIVERABLES — WORKING DEMO MVP

A polished demo showcasing NutriDesh's core AI workflow: scan a Bangladeshi meal and receive a nutrition breakdown; interact with the Bangla AI assistant using meal context and health profile data; explore diabetes-focused nutrition guidance and dashboard tracking. The current prototype demonstrates the product direction, localization strategy, and feasibility of a scalable Bangladesh-first nutrition platform.